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18 February 1965
SUPERSEDING
FA(FC)-PD-98
10 July 1961

MILITARY SPECIFICATION

WRITE SWITCH: 10526581

1. SCOPE

1.1 This specification covers one type of write switch used to select channels for writing information into the memory of the Gun Direction Computer: M18.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts Equipment and Tools for Ordnance Materiel: Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction: M18; Manufacturing Processes and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U.S. Army Munitions Command

F10526581	Assembly, Write Switch

PACKAGING DATA SHEET

M14002	Data Sheet, Master Packaging

FSC 1220

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(Copies of specifications, standards, drawings, and packaging data sheets, required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Fabrication.- The write switch shall be manufactured in accordance with Drawing F10526581 and drawings pertaining thereto.

3.2 General specifications.- The write switch shall meet the following requirements of MIL-F-13926, and MIL-C-46357.

3.2.1 MIL-F-13926

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Workmanship

3.2.2 MIL-C-46357

(a) Etched circuit assembly
(b) Marking
(c) Handling

3.3 Environmental service conditions.

3.3.1 Vibration.- The write switch shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion). The frequency shall be varied uniformly from 10 to 55 cycles per second (cps) and return to 10 cycles. This cycle shall be traversed in 1 minute $\pm$ 15 seconds. The motion shall be applied to the write switch for a period of 10 minutes in each direction of the 3 mutually perpendicular axes (total of 30 minutes), relative to its normal operational position. Subsequent to vibration, the write switch shall not be damaged or adversely affected and shall meet the requirements of 3.4.

3.3.2 Shock.- The write switch shall withstand 3 shock impulses in the vertical plane relative to its normal mounting position. The acceleration of the shock impulses shall be 5 gravity units (G's). Pulse duration of each impulse shall be 11 $\pm$ 1 milliseconds. Subsequent to shock the item shall not have been damaged or adversely affected and shall meet the requirements of 3.4.

3.3.3 Temperatures.

3.3.3.1 Operating.- The write switch shall meet the requirements of 3.4.1.3 while exposed to temperatures of -40°F and +135°F.

3.3.3.2 Storage.- The write switch shall meet the requirements of 3.4 as a temperature of between 60°F and 90°F after having been exposed to each temperature of -80°F and + 160°F.

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3.3.4 Humidity.

3.3.4.1 Operating.- The write switch shall meet the requirements of 3.4.1.3.1 while exposed to an atmosphere with a relative humidity (RH) up to 97%.

3.3.4.2 Non-operating.- The write switch shall meet the requirements of 3.4 after exposure to (and when thoroughly dried) an atmosphere with a RH of 100 percent including condensation, for a period of 4 hours.

3.3.5 Fungus resistance.- The write switch shall show no evidence of damage and shall meet the requirements of 3.4, after being inoculated with the composite spore suspension specified in Specification MIL-F-13926 and incubated at an ambient temperature of between 77°F and 84°F with a RH of 97 ± 2 percent.

3.3.6 Salt spray.- The write switch shall show no evidence of physical damage and meet the requirements of 3.4 immediately after exposure to a fine mist of a 5 percent salt solution concentration. The solution shall have a specific gravity of from 1.0268 to 1.0413 and a pH between 6.5 and 7.2 at a temperature between 93° and 97°F.

3.4 Performance.- The following requirements shall apply for all twelve (12) identical circuits included on the write switch circuit board.

3.4.1 Write switch circuit I (See 6.1).

3.4.1.1 False-state (static conditions).- When a loaded write switch (see Figure 1) is in the false-state (when its input is -0.2 ± 0.2 volt), the following conditions shall exist:

(a) The voltage at the base of the input switch transistor shall be -0.5 ± 0.4 volts (V).
(b) The base-to-emitter voltage of the buffer transistor shall be at least 0.5V, but shall not exceed 1.3V.
(c) The saturation voltage of the buffer transistor shall be at least 0.1V, but shall not exceed 1.2V.
(d) The voltage at the output shall not exceed 25 millivolts.

3.4.1.2 True-state (static conditions).- When a loaded write switch (Figure 2) is in the true-state (its input is -6 ± 1V, the following conditions shall exist:

(a) The voltage at the base of the input transistor shall be -5.5 ± 1.0V.
(b) The emitter to base voltage of the buffer transistor shall be at least 0.4V, but shall not exceed 3.0V.
(c) The saturation voltage of the amplifier transistor shall be at least 0.1V, but shall not exceed 1.2V.

3.4.1.3 Dynamic condition.- With a symmetrical square wave varying between -0.2 ± 0.2V and -6.0 ± 1.0V applied to a loaded write switch input, the output shall be a square wave varying between +0.1 ± 0.1V and +15.0 ± 1.0V.

3.4.1.3.1 Delay time (comparison of input and output waveforms).- The delay time measured from the time the input waveform falls 10 percent of its total excursion from false to true until the output waveform reached 90 percent

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of its final value going towards plus 15V shall be less than 0.5 microsecond (usec). The delay time measured from the time the input waveform reaches 10 percent of its total excursion going from true to false until the output waveform reaches 90 percent of its total excursion from plus 15V to zero shall be less than 0.5 usec (see Figure 3).

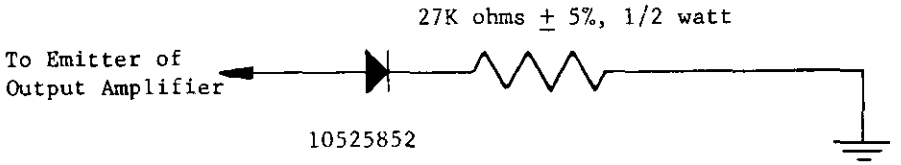


FIGURE 1

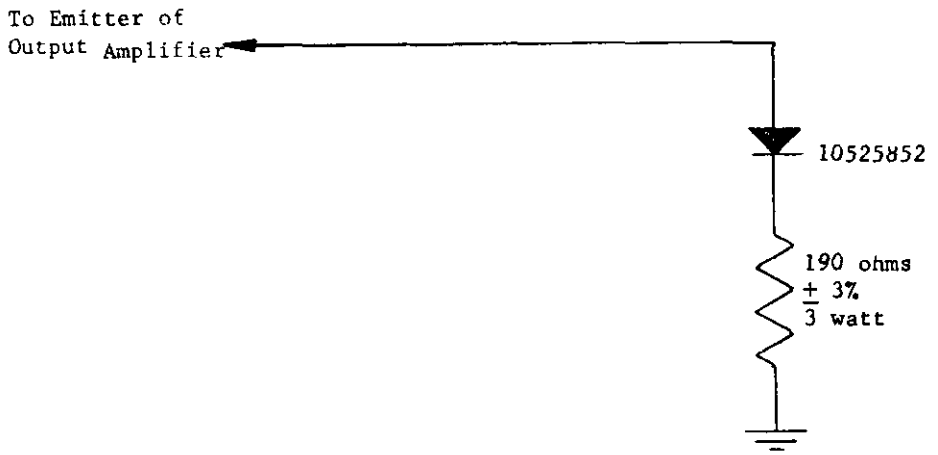


FIGURE 2

Write Switch Load

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3.4.2 Logic diodes.

3.4.2.1 Leakage current.- The leakage current through the logic diodes shall not exceed 20 microamps (ua) when back-biased by 15 ± 2V.

3.4.2.2 Forward voltage drop.- The forward voltage drop across the logic diodes shall not exceed 0.6V at 1.5 milliamperes (ma).

3.4.3 Head isolation diodes.

3.4.3.1 Leakage current.- The leakage current through the head isolation diodes shall not exceed 10 ua when backed-biased by 15V.

3.4.3.2 Forward voltage drop.- The forward voltage drop across the head isolation diodes shall not exceed 1.0V at 80 ± 5 ma.

3.4.4 Test diodes.

3.4.4.1 Leakage current.- The leakage current through the test diodes shall not exceed 0.2 ua when back-biased by 18 ± 2V.

3.4.4.2 Forward voltage drop.- The forward voltage drop across the test diodes shall not exceed 0.8V at 1.5 ma.

3.4.5 Thermistor resistence range.- The resistance range of the thermistor shall be from 6.37 thousand (K) ohms to 16.32K ohms inclusive.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examinations and tests.- Except where otherwise specified herein, components and sub-assemblies shall be inspected in accordance with the "Inspection provisions for components and sub-assemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables I, II, and III and the test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

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4.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.2.4 Functional tests.- The requirements and tests in Table I shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F).

TABLE I

(100 percent inspection)

Characteristic	Requirement	Test Procedures
101. False state (static conditions)	3.4.1.1	4.5.2.1.1
102. True state (static conditions)	3.4.1.2	4.5.2.1.2
103. Dynamic condition	3.4.1.3	4.5.2.1.3
104. Delay time	3.4.1.3.1	4.5.2.1.3.1
105. Leakage current (logic diode)	3.4.2.1	4.5.2.2.1
106. Forward voltage drop (logic diode)	3.4.2.2	4.5.2.2.2
107. Leakage current (head isolation diodes)	3.4.3.1	4.5.2.3.2
108. Forward voltage drop (head isolation diodes)	3.4.3.2	4.5.2.3.1
109. Leakage current (test diodes)	3.4.4.1	4.5.2.4.2
110. Forward voltage drop (test diodes)	3.4.4.2	4.5.2.4.1
111. Thermistor resistance range	3.4.5	4.5.2.5

4.3 Control tests.

4.3.1 General.- One write switch shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table II.

TABLE II

CONTROL TEST	REQUIREMENT	TEST PROCEDURES
301. Fabrication	3.1	Applicable drawings - visual
302. General specifications	3.2	MIL-F-13926 and MIL-C-46357 visual

4.3.2 Environmental.- Three units shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall have met the requirements and tests in Table I and shall then meet the requirements and tests in Table III.

TABLE III

CONTROL TEST	REQUIREMENT	TEST PROCEDURES
303. Vibration	3.3.1	4.5.1.1
304. Shock	3.3.2	4.5.1.2
305. Humidity-Operating	3.3.4.1, 3.4.1.3.1	4.5.1.3.2, 4.5.2.1.3.1

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TABLE III (Cond)

CONTROL TEST	REQUIREMENT	TEST PROCEDURES
306. Humidity -non-operating	3.3.4.2	4.5.1.3.1
307. Salt spray	3.3.6	4.5.1.4
308. Operating temperature	3.3.3.1, 3.4.1.3	4.5.1.5, 4.5.2.1.3
309. Storage temperature	3.3.3.2	4.5.1.5

4.3.3 Control test failure.- Should any one item of a control test sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.1.1 Test equipment.- In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5:

Item Number	Description
1. Oscilloscope	Dual channel, rise time of scope plus associated measuring probe 50 millimicroseconds or less, sweep period 50 millimicroseconds per centimeter or greater, input resistance greater than 8 megohms, input capacitance less than 25 micro-microfarads.
2. Generator; Square Wave	Shall have a fundamental frequency of 250 +25kc to supply a symmetrical square wave from minus 0.15 ± 0.15 volts to minus 6 ± 1 volt, with rise and fall times each less than or equal to 0.2 microseconds, output impedance 100 ohms or less.

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Item Number	Description
3. Voltage measuring device; d.c.	Accuracy ± 5 percent, voltage range 10 mv to greater than 100 volts. Input resistance greater than or equal to 100K ohm
4. Power supply	Plus 35 volts ± 2 percent, current rating 300 milliamps or greater, peak-to-peak ripple less than or equal to 100 millivolts, output impedance less than 2 ohms from D.C. to 500kc.
5. Power supply	Plus 15 volts ± 2 percent, current rating 350 milliamps or greater, peak-to-peak ripple less than or equal to 100 millivolts, output impedance less than 2 ohms from D.C. to 500kc.
6. Power Supply	Minus 18 volts ± 2 percent, current rating 50 milliamps or greater, peak-to-peak ripple less than or equal to 100 millivolts, output impedance less than 2 ohms from D.C. to 500kc.
7. Power supply	Minus 6 ± 2 percent, current rating 200 milliamps or greater, peak-to-peak ripple less than or equal to 100 millivolts, output impedance less than 2 ohms from D.C. to 500kc.
8. Power supply	Minus 3 volts ± 2 percent, current rating 300 milliamps or greater, peak-to-peak ripple less than or equal to 100 millivolts, output impedance less than 2 ohms from D.C. to 500kc.
9. Resistor	27k ohm ± 5% 1/2 W
10. Resistor	190 ohm ± 3% 3 W
11. Resistor (4)	220 ohm ± 5% 3W
12. Resistor (5)	10k ohm ± 5% 1/2 W
13. Resistor	100k ohm ± 5% 1/2 W
14. Resistor	82k ohm ± 5% 1/8 W

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4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 Vibration.- The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.3.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.3.1 with the write switch rigidly mounted in its normal operating position. Subsequent to the vibration test, the write switch shall show no damage or defect resulting from the test when examined by visual and tactile means. The write switch shall then be inspected as specified in 4.5.1.2 prior to determining compliance with requirements specified under 3.4.

4.5.1.2 Shock.- This test shall be conducted in accordance with Procedure II of the shock test defined in MIL-F-13926 except that the write switch shall be shocked in the vertical plane only. The impact (shocks) and "G" value shall be in accordance with 3.3.2 A receiving device, representing the connector receptacle that receives the write switch in the computer, shall be secured to the horizontal surface of the shock tester. Each end of the receiving device shall be clamped such that the female receiving slot will hold the unit in a vertical plane. The unit will then be inserted into the receiving device and shocked. Subsequent to imposing the number of impacts required, a visual examination shall be made for damages or defects to determine compliance with 3.3.2 and then the unit will be inspected as specified in 4.5.1.3 prior to determining compliance with 3.4 as required by 3.3.2.

4.5.1.3 Humidity.

4.5.1.3.1 Non-operating.- This test shall be conducted in accordance with the conditions in 3.3.4.2 and the humidity test defined in Procedure I of MIL-F-13926 except that the 4-1/2 hour period shall be reduced to 2 hours, the 6 hour period reduced to 2 hours, and the test period shall not exceed 5.5 hours per cycle. The unit shall then be thoroughly dried and then subjected to the test in 4.5.1.3.2 prior to meeting the specified requirements of 3.4 as required by 3.3.4.2.

4.5.1.3.2 Operating.- This test shall be conducted in accordance with the conditions in 3.3.4.1 and the humidity test defined in Procedure III of MIL-F-13926 except that the humidity shall be maintained at 97 $\pm$ 5 percent for the 24 hour period, and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). While exposed to this test condition the write switch shall be subjected to the test specified in 4.5.2.1.3.1 and shall meet the requirements referenced therein to determine compliance with 3.3.4.1.

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4.5.1.4 Salt spray. - The Write Switch shall be subjected to the salt spray test defined in MIL-STD-202, Condition B. Immediately after exposure, the Write Switch shall be inspected for any physical damage, then tested in accordance with 4.5.2 for compliance with 3.3.6.

4.5.1.5 Temperature (Operating and Storage). - After being subjected to all environmental tests specified above, the Write Switch shall be subjected to Test Procedure I specified in MIL-F-13926, except the temperature specified in 3.3.3 shall apply. While at the operating temperatures of 3.3.3.1, the Write Switch shall be tested in accordance with 4.5.2.1.3 to determine compliance with the requirements of 3.4.1.3 as specified in 3.3.3.1. The temperature shall then be returned to standard temperature (60° to 90°F) and the Write Switch shall be tested in accordance with 4.5.2 to 4.5.2.5 inclusive to determine compliance with 3.3.3.2.

4.5.1.6 Fungus. - To determine compliance with 3.3.5, one of the first 5 Write Switches produced shall be submitted for testing in accordance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.

4.5.2 Performance. - To determine compliance with 3.4 the 12 Write Switch circuits on the module shall each be subjected to the tests specified in 4.5.2.1 to 4.5.2.1.3.1 inclusive and shall meet the requirements referenced therein when the corresponding pins and test points as shown in Table IV are respectively compared. Column 1 of Table IV shows the pins and test points of Write Switch 1 in the order given in the tests from 4.5.2.1 to 4.5.2.1.3.1 inclusive. Each of the 11 following columns of Table IV show, in the same order, corresponding pins and test points for the 11 remaining Write Switch circuits. Load only 1 Write Switch at a time.

4.5.2.1 Write Switch Circuit I. - Connect the power supplies to the respective pins as specified in Table V. The D.C. returns of each of the power supplies used for testing this circuit shall be connected to pin 35A.

TABLE V

Pin	Power Supply
60A	+15 V (Item 5, par. 4.4.1.1)
50A	- 6 V (Item 7, par. 4.4.1.1)
17A	-18 V (Item 6, par. 4.4.1.1)
52A	- 3 V (Item 8, par. 4.4.1.1)

4.5.2.1.1 False state (static conditions) tests. - Connect the +35 volt power supply (item 4 of 4.4.1.1) to pin 13A. Connect the +15 volt power supply (item 5 of 4.4.1.1) to pin 7B and connect a 27k ohm resistor (Item 9 of 4.4.1.1) from pin 8B through a 10525852 diode to pin 35A. (See Figure 1). With this condition established connect pin 5B to power ground (pin 35A) and determine the following in compliance with 3.4.1.1 (a) to (d) inclusive:

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Table IV-Corresponding Pins and Test Points

WRITE SWITCH											
1	2	3	4	5	6	7	8	9	10	11	12
Pin	Pin	Pin	Pin	Pin	Pin	Pin	Pin	Pin	Pin	Pin	Pin
13A	41B	13A	41B	13A	59B	13A	51B	6B	62A	1B	62A
7B	44B	11B	50B	14B	55B	20B	57B	1A	57A	5A	63B
8B	45B	12B	52B	15B	56B	22B	60B	2A	58A	6A	64B
35A	35A	35A	35A	35A	35A	35A	35A	35	35A	35A	35A
5B	44A	10A	54A	12A	54B	22A	61B	3A	55A	11A	64A
8A	49B	15A	48B	18B	46B	19B	56A	2B	58B	7A	65B
TP3	TP6	TP9	TP12	TP15	TP18	TP21	TP24	TP27	TP30	TP33	TP36
TP1	TP2	TP7	TP8	TP13	TP14	TP19	TP20	TP26	TP25	TP32	TP31

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(a) The voltage from pin 8A to pin 35A shall be minus 0.25 ± 0.25 volts.

(b) The voltage from TP3 to the minus 3 volt supply shall be plus 0.9 ± 0.4 volts.

(c) The voltage at TP1 to the minus 3 volt supply shall be plus 0.65 ± 0.55 volt.

(d) The voltage at the output (pin 8B) to pin 35A shall not exceed 25 millivolts.

4.5.2.1.2 True state tests.- Disconnect the 27k resistor from pin 8B and the 10525852 diode. Connect a 190 ohm resistor (Item 10 of 4.4.1.1) from pin 8B to the 10525852 diode, which is connected to pin 35A. (See Figure 2). With this condition established connect pin 5B to the minus 6 volt supply and determine the following in compliance with 3.4.1.2(a) to (c) inclusive:

(a) The voltage at pin 8A to pin 35A shall be minus 5.5 ± 1.0 volts.

(b) The voltage from TP3 to the minus 3 volt supply shall be minus 1.7 ± 1.3 volts.

(c) The voltage from pin 7B to pin 8B shall be plus 0.65 ± 0.55 volt.

4.5.2.1.3 Dynamic conditions testing.- Disconnect the test lead connected to pin 5B and apply an input test signal to pin 5B of write switch I by means of a signal generator (Item 2 of 4.4.1.1). The standard input signal to be used for all dynamic tests for the write switch shall be an approximately symmetrical square wave (each half period shall be 2.00 ± 0.1 microseconds) varying from minus 0.2 ± 0.2 volts to minus 6.0 ± 1.0 volt, with 10 to 90 percent rise and fall times each less than 0.4 microseconds. The output waveform (Pin 8B) as measured on a dual channel oscilloscope (Item I of 4.4.1.1) shall be a square wave varying between the levels of plus 0.1 ± 0.1 volt to plus 15.0 ± 1.0 volts at the frequency of the input signal in compliance with 3.4.1.3.

4.5.2.1.3.1 Delay time (comparison of input and output waveforms).- The input test signal specified in 4.5.2.1.3 shall be compared with the output wavefore on a dual channel oscilloscope (Item I of 4.4.1.1). The delay time measured from the time the input test signal falls 10 percent of its total excursion (going from zero to minus 6 volts) until the output reaches 90 percent of its final value (going towards plus 15 volts) shall be less than 0.5 microseconds. The delay time (See Figure 3) measured from the time the input test signal reaches 10 percent of its total excursion (going from minus 6 to zero volts) until the output reaches 90 percent of its total excursion (going from plus 15 to zero volts) shall be less than 0.5 microseconds in compliance with 3.4.1.3.1.

4.5.2.2 Logic diodes tests.- Remove all external connections including voltage levels and power ground. Only those voltages specified shall be applied to the module.

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4.5.2.2.1 Leakage current. - Apply the plus 15 volt supply (item 5 of 4.4.1.1) to pin 17A. Ground the pin specified in Column A of Table VI while reading the voltage from the plus 15 volt supply to the corresponding pin specified in Column B of Table VI. This voltage shall not exceed 240 millivolts to determine conformance with the requirements of 3.4.2.1.

TABLE VI

Column A		Column B
Diodes	Pin to Ground	Pin to +15 volt supply
CR-9	10A	15A
CR-7	10A	8A
CR-11	5B	15A
CR-5	5B	8A
CR-21	12A	18B
CR-27	12A	19B
CR-23	22A	18B
CR-25	22A	19B
CR-37	3A	2B
CR-39	4B	2B
CR-41	11A	7A
CR-43	4A	7A
CR-6	44A	49B
CR-12	44A	48B
CR-8	54A	49B
CR-10	54A	48B
CR-22	54B	46B
CR-28	54B	56A
CR-24	61B	46B
CR-26	61B	56A
CR-38	55A	58B
CR-40	62B	58B
CR-42	64A	65B
CR-44	63A	65B

4.5.2.2.2 Forward voltage drop.- Remove the plus 15 volt power supply specified in 4.5.2.2.1 from pin 17A. Connect the minus 18 volt power supply (Item 6, par. 4.4.1.1) to pin 17A. Ground the pin specified in column A of Table VII while reading the voltage from the pin specified in column B of Table VII to ground. This voltage shall be minus 0.30 ± 0.30 volts to determine conformance with 3.4.2.2.

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TABLE VII

Column A		Column B
Diode	Pin to Ground	Pin to Ground
CR-11	5B	15A
CR-7	10A	8A
CR-27	12A	19B
CR-23	22A	18B
CR-39	4B	2B
CR-43	4A	7A
CR-12	44A	48B
CR-8	54A	49B
CR-28	54B	56A
CR-24	61B	46B
CR-40	62B	58B
CR-44	63A	65B

4.5.2.3 Head isolation diodes

4.5.2.3.1 Forward voltage drop. - Connect a 220 ohm resistor (Item 11, par. 4.4.1.1) to each of the following pins from the minus 18 volt power supply (Item 6, par 4.4.1.1) 33B, 34A, 32B, 32A. Ground the pin specified in column A of Table VIII while reading the voltage from the pin specified in column B to ground. This voltage shall be minus 0.5 ± 0.5 volts to determine conformance with 3.4.3.2.

TABLE VIII

Column A		Column B
Diode	Pin	Pin
CR-49	31B	33B
CR-50	31A	33B
CR-51	30B	33B
CR-52	30A	33B
CR-53	34B	33B
CR-54	35B	33B
CR-55	36A	33B
CR-56	36B	33B
CR-57	29B	34A
CR-58	29A	34A
CR-59	28B	34A
CR-60	28A	34A
CR-61	37A	34A
CR-62	37B	34A
CR-63	38A	34A
CR-64	38B	34A
CR-65	26B	32B
CR-66	26A	32B
CR-67	19A	32B

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TABLE VIII (Cont)

Column A		Column B
Diode	Pin	Pin
CR-68	20A	32B
CR-69	39A	32B
CR-70	48A	32B
CR-71	46A	32B
CR-72	45A	32B
CR-73	23A	32A
CR-74	23B	32A
CR-75	24A	32A
CR-76	24B	32A
CR-77	43A	32A
CR-78	43B	32A
CR-79	41A	32A
CR-80	40B	32A

4.5.2.3.2 Leakage current.- Remove the 220 ohm resistors from the pins specified in 4.5.2.3.1. Connect 10K ohm resistors (Item 12, paragraph 4.4.1.1) to each of the following pins from the plus 15 volt supply (Item 5, paragraph 4.4.1.1) 33B, 34B, 32B, 32A. Ground the pin specified in Column A of Table VIII while reading the voltage from the plus 15 olt power supply to the pin specified in Column B of Table VI. This voltage shall not exceed 100 millivolts in determining compliance with the requirements of 3.4.3.1.

4.5.2.4 Test diodes.

4.5.2.4.1 Forward voltage drop.- Connect a 10K ohm resistor (Item 12, paragraph 4.4.1.1) from pin 33A to the plus 15 volt power supply (Item 5, paragraph 4.4.1.1). Ground each pin specified in Column A of Table IX in sequence while reading the voltages from pin 33A to ground. The voltage shall not exceed the voltage specified in 3.4.4.2.

TABLE IX

Column A		Column B
Diode	Pin to Ground	Pin to +15 volt supply
CR-81	25B	33A
CR-82	25A	33A
CR-83	40A	33A
CR-84	39B	33A

4.5.2.4.2 Leakage current.- Remove the plus 15 volt power supply and the 10K ohm resistor specified in 4.5.2.4.1 from pin 33A. Connect the minus 18 volt power supply (Item 6, paragraph 4.4.1.1) through a 100K ohm resistor (Item 13, paragraph 4.4.1.1) to pin 33A. Ground each pin specified in Column A of Table IX in sequence while reading the voltages from pin 33A to ground. The voltage shall not exceed 20 millivolts in determining compliance with the requirements of 3.4.4.1.

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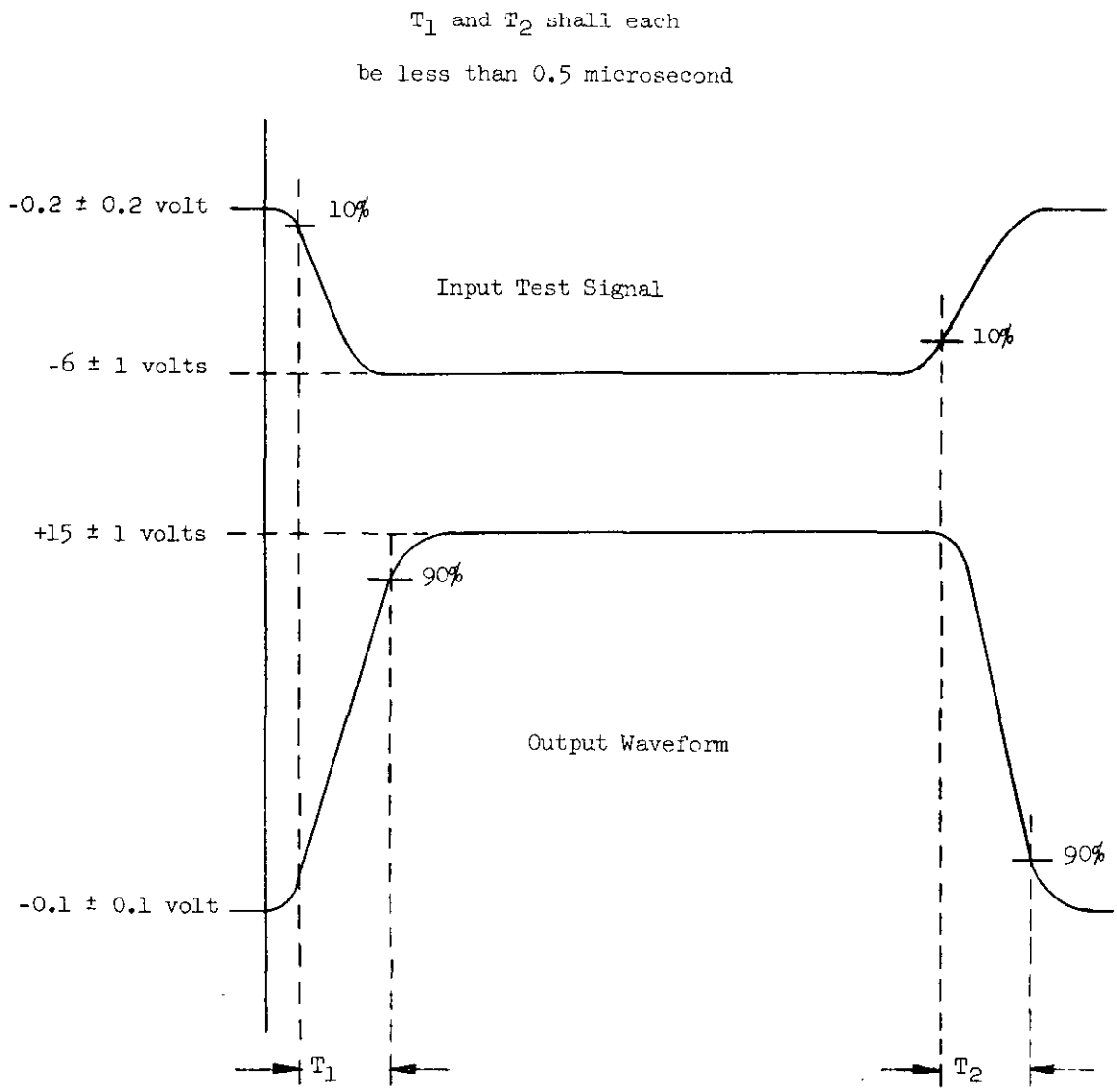


FIGURE 3

Dynamic Tests - Write Switch

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4.5.2.5 Thermistor resistor range. - Connect an 82K ohm resistor (Item 14, par. 4.4.1.1) from the minus 6 volt supply (Item 7, par. 4.4.1.1) to pin 10B. Connect the D.C. return of the minus 6 volt supply to pin 13B. The voltage from pin 10B to pin 13B shall be minus 0.77 ± 0.33 volts (See Figure 4) in determining compliance with the requirements of 3.4.5.

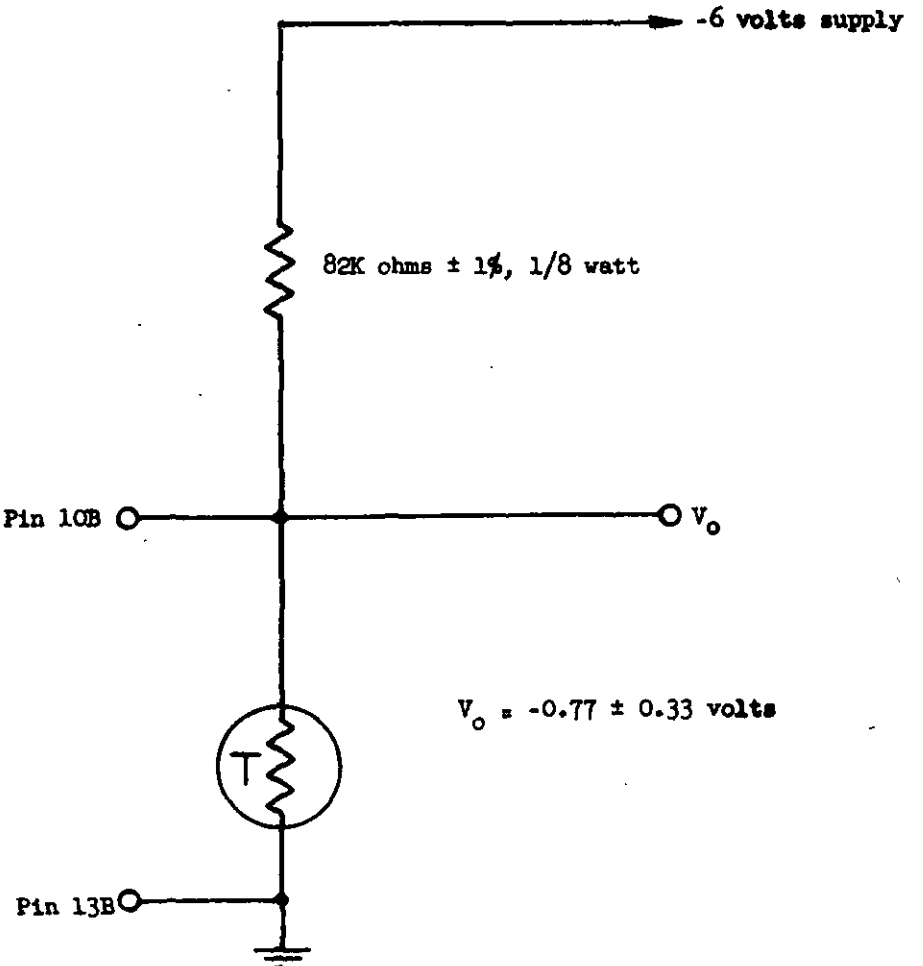


FIGURE 4

Checking Thermistor Resistance

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing.- Preservation, packaging and packing shall be in accordance with Master Packaging Sheet M14002.

6. NOTES

6.1 Intended use.- The Write Switch module contains 12 separate Write Switches, write-head isolation diodes, test diodes, and a temperature sensing thermistor. The Write Switches are used to complete continuity from the center-taps of main memory write-heads to selected write amplifiers. The head isolation diodes isolate the selected write-heads from other write-heads in the Write Switch Matrix. The test diodes are used for logic testing circuitry elsewhere in the computer. The thermistor is used to monitor the temperature of the Write Switch module; this thermistor is connected to power control circuitry elsewhere in the computer.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of applicable level of preservation, packaging and packing
(c) Applicable stock number
(d) Applicable packaging data sheet (see 5.1)
(e) A block of serial numbers to identify each write switch

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1220-A177Y

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 OCT 64 1426

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